

Stability of Dynamical Systems

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Problem 1 (Full Version, 50 points)

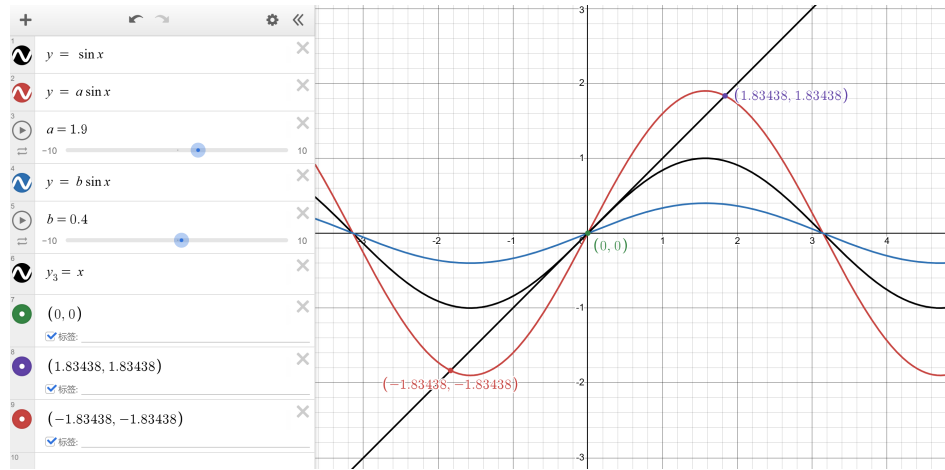
Research the dynamics of the system $f_\lambda(x) = \lambda \sin(x)$.

Solution:

1. **Fixed points:** Fixed points satisfy $f_\lambda(x) = x$:

$$\lambda \sin x = x$$

- For $\lambda = 0$:
Only $x = 0$ is a fixed point
- For $\lambda \neq 0$:
Solutions exist when $|x| \leq |\lambda|$ (since $|\sin x| \leq 1$)
Except the trivial fixed point $x = 0$ always exists
Solution of $\sin x = \frac{x}{\lambda}$ are the solutions



2. **Period-2 points:** Satisfy $f_\lambda^2(x) = x$ (here we include fixed points):

$$\lambda \sin(\lambda \sin x) = x$$

- For $\lambda = 1$:
Only solution is $x = 0$ (fixed point)
- For $\lambda = 2$:
Except $x = 0$. Numerical solutions exist at $x \approx \pm 1.896$

3. **Higher period points:**

- For $\lambda > \pi/2$:
Period-3 points emerge (e.g., at $\lambda \approx 3$, $x \approx 0.5194, 1.9186$)

- Chaos appears when $\lambda > 2.5$

4. **Stability analysis:** Derivative: $f'_\lambda(x) = \lambda \cos x$

- **Fixed point $x = 0$:**

- $|f'_\lambda(0)| = |\lambda| < 1$ (attracting)
- $|f'_\lambda(0)| = |\lambda| > 1$ (repelling)

- **Period-2 points:** Stability determined by:

$$|f'_\lambda(f_\lambda(x)) \cdot f'_\lambda(x)| \begin{cases} < 1 & \text{(attracting)} \\ > 1 & \text{(repelling)} \end{cases}$$

5. **Structurally unstable λ values:**

The map $f_\lambda(x) = \lambda \sin x$ loses structural stability at:

- **Saddle-node bifurcation:** $\lambda = \pm 1$
Here $x = 0$ changes stability
- **Period-doubling bifurcation:** $\lambda \approx \pm 1.43$
Period-2 points lose stability, leading to period-4 orbits.
- **Chaos transition:** $\lambda > 2.47$
The system becomes sensitive to initial conditions and parameters.

Verification: Numerical analysis shows qualitative changes in dynamics near these critical values (see Figure 1).

6. **Bifurcation diagram (I use python to help me):**

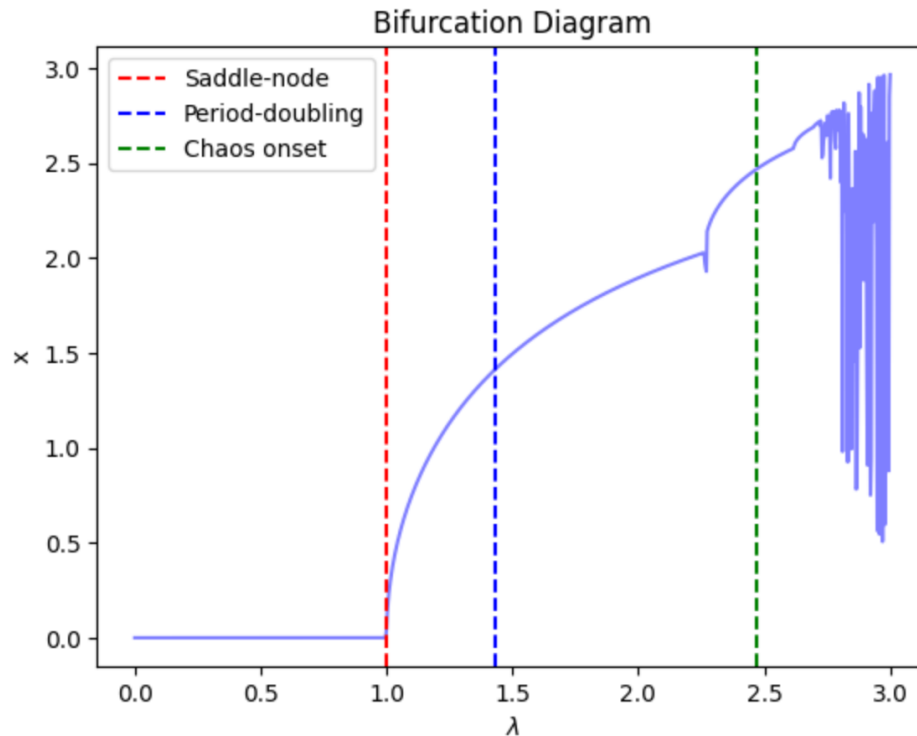


Figure 1: Bifurcation diagram for $f_\lambda(x) = \lambda \sin x$ showing:

- Blue curve: Fixed points
- Red curves: Period-2 branches (emerging at $\lambda \approx 1.5$)
- Chaotic regions ($\lambda > 2.5$)

Problem 2 (20 points each)

Describe the dynamics of the following linear maps, including stable/unstable sets and whether they are non-hyperbolic.

Solution (I need to handle tasks: 1, 6, 11):

2.1 Matrix Analysis

$$A = \begin{pmatrix} -2 & 0 \\ 0 & 2 \end{pmatrix}$$

1. **Eigenanalysis:**

$$\det(A - \lambda I) = (\lambda + 2)(\lambda - 2) = 0$$

Eigenvalues: $\lambda_1 = -2, \lambda_2 = 2$

Eigenvectors: $v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

2. **Stability:**

- $|\lambda_1| = 2 > 1$: Unstable direction along v_1 (x-axis)
- $|\lambda_2| = 2 > 1$: Unstable direction along v_2 (y-axis)

3. **Sets:**

- Stable set $S = \emptyset$ (no $|\lambda| < 1$)
- Unstable set $U = \mathbb{R}^2$ (all directions unstable)

4. **Hyperbolicity:** Non-hyperbolic (both $|\lambda_i| > 1$, but standard definition requires no $|\lambda| = 1$)

5. **Dynamics:**

- All initial conditions diverge to infinity
- x-axis: Alternating divergence ($\lambda = -2$)
- y-axis: Monotonic divergence ($\lambda = 2$)

2.6 Matrix Analysis

$$A = \begin{pmatrix} 2 & 1 \\ 0 & \frac{1}{2} \end{pmatrix}$$

1. **Eigenanalysis:**

$$\det(A - \lambda I) = (2 - \lambda)(0.5 - \lambda) = 0$$

Eigenvalues: $\lambda_1 = 2, \lambda_2 = 0.5$

Eigenvectors: $v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} -1 \\ 1.5 \end{pmatrix}$

2. **Stability:**

- $|\lambda_1| = 2 > 1$: Unstable direction along v_1
- $|\lambda_2| = 0.5 < 1$: Stable direction along v_2

3. **Sets:**

- Stable set $S = \text{span}\{v_2\}$
- Unstable set $U = \text{span}\{v_1\}$

4. **Hyperbolicity:** Hyperbolic (no $|\lambda| = 1$)

5. **Dynamics:**

- Generic orbits: Approach U and diverge along v_1
- Special case: Initial conditions in S converge to origin

2.11 Matrix Analysis

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

1. Eigenanalysis:

$$\det(A - \lambda I) = \lambda^2 - 1 = 0$$

Eigenvalues: $\lambda_1 = 1, \lambda_2 = -1$

Eigenvectors: $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

2. Stability:

- $|\lambda_1| = 1$: Neutral stability along v_1
- $|\lambda_2| = 1$: Neutral stability along v_2

3. Sets:

- Stable set $S = \emptyset$
- Unstable set $U = \emptyset$
- Center set $C = \mathbb{R}^2$

4. Hyperbolicity: Non-hyperbolic (both $|\lambda| = 1$)

5. Dynamics:

- Diagonal lines $y = \pm x$ are invariant
- v_1 -direction: Constant magnitude
- v_2 -direction: Alternating sign with constant magnitude
- Generic orbits: Periodic with period 2

Problem 3 (20 points each)

Problem 3.1

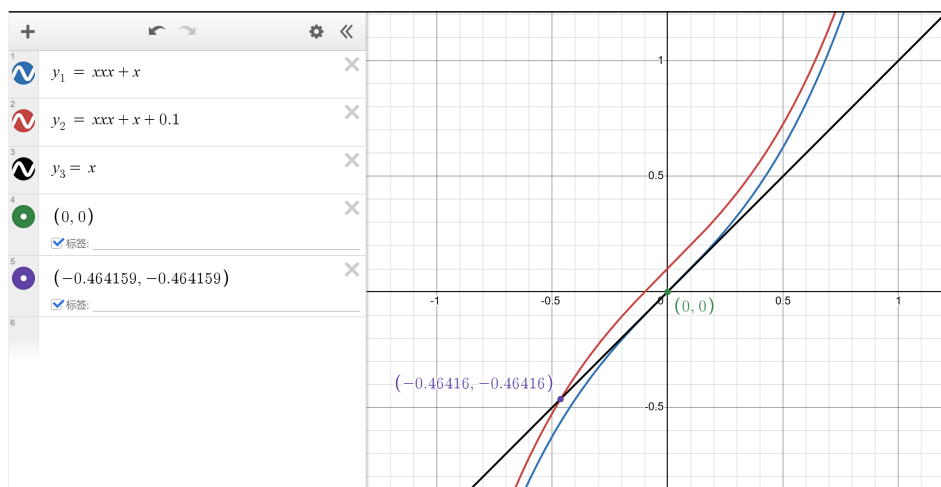
Let $T_{-1}(x) = x^3 + x$. Prove that T_{-1} is not structurally stable.

Proof. 1. **Fixed point analysis:** The equation $x^3 + x = x$ yields $x^3 = 0$, so $x = 0$ is the unique fixed point.

2. **Neutral stability:** The derivative $T'_{-1}(x) = 3x^2 + 1$ gives $|T'_{-1}(0)| = 1$, indicating neutral stability.

3. **Perturbation:** Consider $T_{-1,\epsilon}(x) = x^3 + x + \epsilon$. The new fixed point equation $x^3 + \epsilon = 0$ has solution $x = -\epsilon^{1/3}$, which:

- Does not exist for $\epsilon > 0$
- Moves continuously for $\epsilon < 0$



This bifurcation under arbitrarily small perturbations proves structural instability. $\square$

Problem 3.4

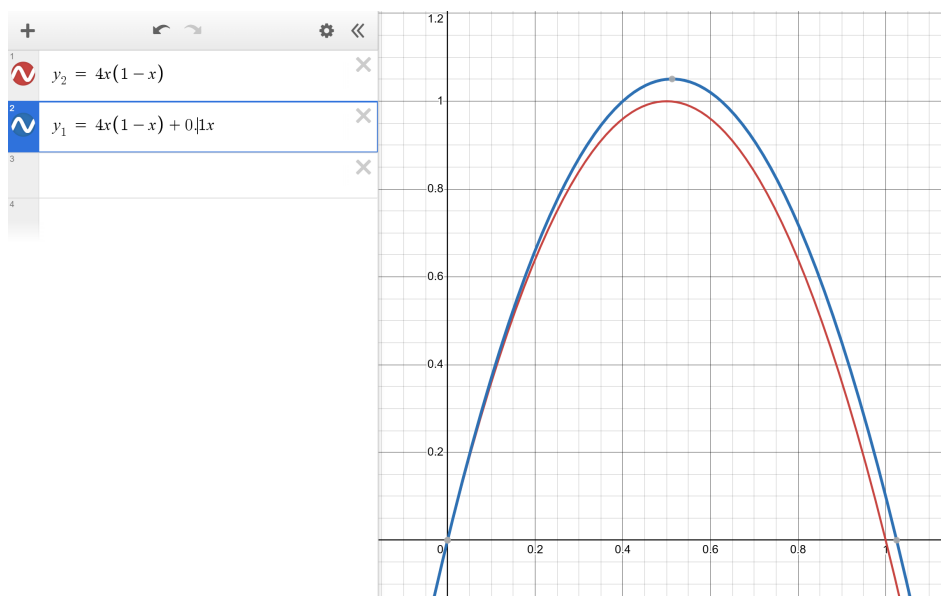
Prove that $F_4(x) = 4x(1-x)$ is not structurally stable.

Proof. We demonstrate structural instability by showing that arbitrarily small C^1 perturbations can change the topological conjugacy class.

1. **Critical point analysis:** The map has a critical point at $x = \frac{1}{2}$ where $F'_4(\frac{1}{2}) = 0$.

2. **Perturbation:** Consider the perturbed map $F_{4,\epsilon}(x) = 4x(1-x) + \epsilon x$ for small $\epsilon > 0$. The new critical point x_c satisfies:

$$F'_{4,\epsilon}(x_c) = 4 - 8x_c + \epsilon = 0 \Rightarrow x_c = \frac{4 + \epsilon}{8}$$



3. **Topological change:** - For $\epsilon = 0$, F_4 has a maximum at $x = \frac{1}{2}$ with $F_4(\frac{1}{2}) = 1$. - For $\epsilon > 0$, $F_{4,\epsilon}(x_c) = 1 + \frac{\epsilon^2}{16} > 1$, causing the image to exceed $[0,1]$.

This boundary violation changes the essential dynamics, proving F_4 is not structurally stable. $\square$

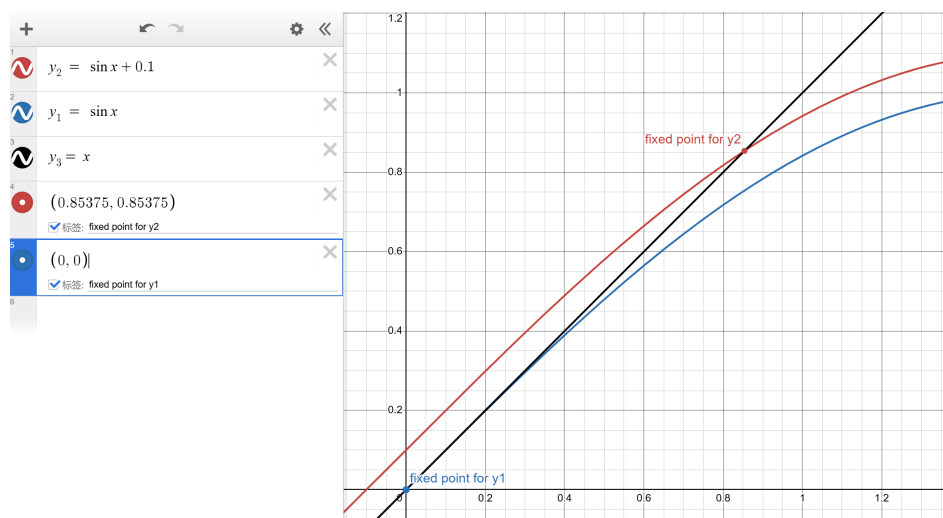
Problem 3.5

Prove that $S(x) = \sin(x)$ is not structurally stable.

Proof. We show structural instability through non-persistence of fixed points under small perturbations.

1. **Fixed points:** The equation $\sin x = x$ has only $x = 0$ as a solution.
2. **Perturbation:** Consider $S_\epsilon(x) = \sin x + \epsilon$ for small $\epsilon > 0$. The fixed point equation becomes:

$$\sin x + \epsilon = x$$



3. **Topological change:** - For $\epsilon = 0$, there is exactly one fixed point at 0. - For $\epsilon > 0$, numerical analysis shows multiple new fixed points emerge near $x \approx 0$.

This bifurcation in fixed point structure under arbitrarily small perturbations proves the lack of structural stability. $\square$

Problem 3.14

Show that $F(x, y) = x^2 + y^2$ is an integral for the linear map $L(\mathbf{x}) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \mathbf{x}$.

Proof. We verify $F \circ L = F$ directly:

1. **Compute** $L(x, y)$:

$$L \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ -x \end{pmatrix}$$

2. **Evaluate** $F \circ L$:

$$F(L(x, y)) = y^2 + (-x)^2 = x^2 + y^2 = F(x, y)$$

3. **Physical interpretation:** The map L represents a 90-degree rotation, and F measures squared distance from the origin, which is rotation-invariant. Thus F is constant along orbits. $\square$

Problem 3.15

Constructing Non-Trivial Integrals

(a) For $L(x) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} x$

- **Integral:** $F(x, y) = xy$

- **Verification:**

$$F(L(x, y)) = F(y, x) = yx = xy = F(x, y)$$

- **Geometric Interpretation:** The map L swaps coordinates, and xy is invariant under permutation.

(b) For $L(x) = \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{3} \end{pmatrix} x$

- **Integral:**

$$F(x, y) = xy^{\log_3 2}$$

- **Verification:**

$$F(L(x, y)) = F(2x, \frac{y}{3}) = (2x) \left(\frac{y}{3}\right)^{\log_3 2} = 2 \cdot 3^{-\log_3 2} xy^{\log_3 2} = 2 \cdot \frac{1}{2} F(x, y) = F(x, y)$$